Descriptive Statistics - Analytical Challenge Problems

Wednesday, September 27, 2017 5:54 PM

$R_A \sim N(\mu_A, \sigma_A^2)$ $\qquad \mu, \sigma^2$

$R_G \sim N(\mu_G, \sigma_G^2)$

$w_A \Rightarrow R_A$

$w_G \Rightarrow R_G$

$$E[R_P] = w_A E[R_A] + w_G E[R_G] \qquad \text{Mean}$$
$$= w_A \cdot \mu_A + w_G \cdot \mu_G$$

Mean, Average, Expected Value] Same

$$Var(R_P) = \sigma_P^2 = w_A^2 \cdot \sigma_A^2 + w_G^2 \cdot \sigma_G^2 + 2 \cdot w_A \cdot w_G \cdot \sigma_{AG}$$

Variance

$$Cov(R_A, R_G) = \sigma_{AG}$$

$$r_{AG} = \boxed{\frac{\sigma_{AG}}{\sigma_A \cdot \sigma_G}} \qquad \text{Correlation}$$

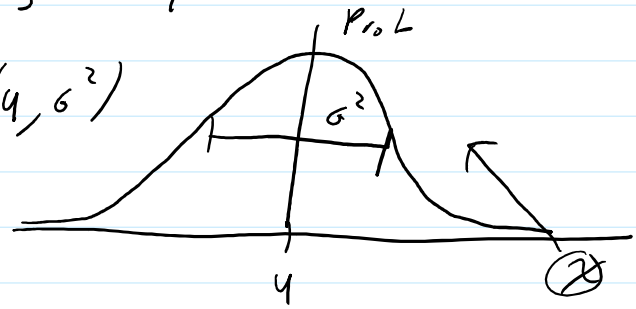
$$\sigma_{AG} = r_{AG} \cdot \sigma_A \cdot \sigma_G \qquad \text{Covariance}$$

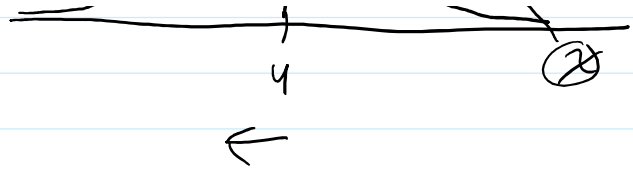
$$RP = R - R_f \qquad \text{Sharpe} = \frac{E[R] - R_f}{\sigma} = \frac{RP}{\sigma}$$

$$RP_A = E[R_A] - R_f \qquad \text{Sharpe}_A = \frac{E[R_A] - R_f}{\sigma_A} = \frac{RP_A}{\sigma_A}$$

$$RP_G = E[R_G] - R_f$$

Normal $x \sim N(\mu, \sigma^2)$





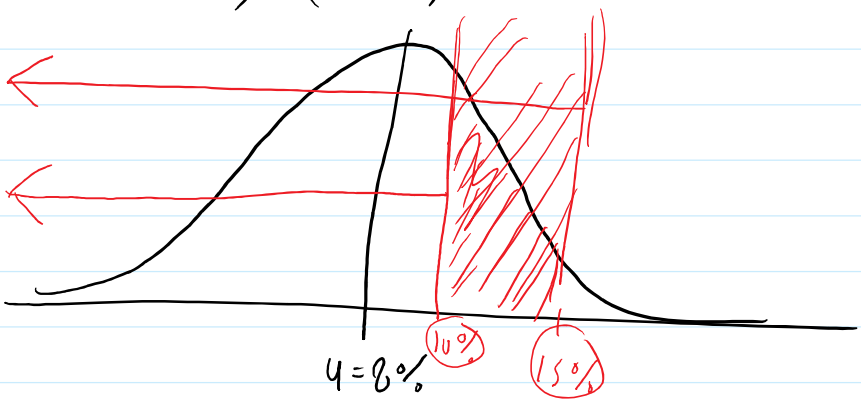
$y = \text{norm.dist}(x)$
$x = \text{norm.inv}(y)$

$R_f = 5\%$ $\qquad RP_A = 3\% = \underbrace{E[R_A]}_{8\%} - \underbrace{R_f}_{5\%}$

$\Rightarrow E[R_A] = 8\%$

$\text{Sharpe} = 0.5 = \dfrac{RP_A}{\sigma_A} = \dfrac{3\%}{\sigma_A} \Rightarrow \sigma_A = \dfrac{3\%}{0.5} = 6\%$

$R_A \sim N(8\%, (6\%)^2)$



1. $\mu, \sigma \Rightarrow$ norm.dist( )

Prob 2 $\qquad E[R_A] = R_A$

$\text{Sharpe}_A = \boxed{0.5} = \dfrac{R_A - R_f}{\sigma_A} = \boxed{\dfrac{6\%}{\sigma_A}} \Rightarrow \sigma_A = \dfrac{6\%}{.5} = .12$

$\text{Sharpe}_G = 0.7 = \dfrac{R_G - R_f}{\sigma_G} = \dfrac{8\%}{\sigma_G} \Rightarrow \sigma_G = \dfrac{8\%}{0.7}$

$$\text{Sharpe}_G = 0.7 = \frac{R_G - R_T}{\sigma_G} = \frac{0.08}{\sigma_G} \rightarrow \sigma_G = \frac{0.08}{0.7} = .11$$

$$r_{AG} = 0.8 = \frac{\sigma_{AG}}{\sigma_A \sigma_G} \qquad \sigma_{AG} = 0.8 \times 0.12 \times 0.11 = .0110$$

$RA_A = 6\%$ $\qquad RP_G = 8\%$

Problem 3

$$R_P = 6.2\% = w_A R_A + w_G R_G = 60\% \cdot 5\% + 40\% \cdot R_G$$

$R_A = 5\%$

$$\Rightarrow R_G = \frac{6.2\% - 60\% \cdot 5\%}{40\%} = 8\%$$

$w_A + w_G = 1$

$w_G = 1 - w_A = 1 - 60\% = 40\%$

Problem 4

$R_A = 5\%$ $\qquad \sigma_A^2 = 0.0144$ $\qquad r = -0.5$

$R_G = 6\%$ $\qquad \sigma_G^2 = 0.0100$

$w_A = 20\%$ $\qquad \sigma_{AG} = r \cdot \sigma_A \cdot \sigma_G = -0.5 \cdot \sqrt{0.0144} \cdot \sqrt{0.0100} = -0.006$

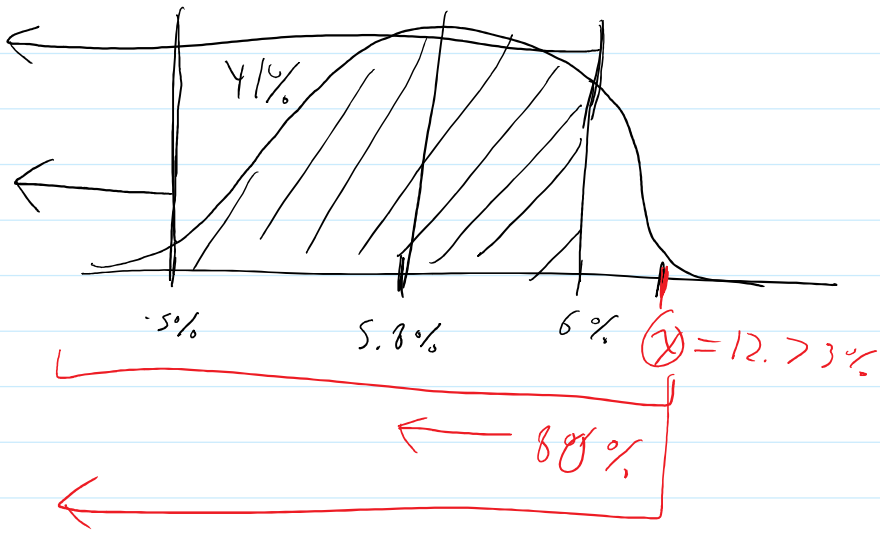
$w_G = 80\%$

$E[R_P] = \mu_P$ $\qquad \sigma_P$

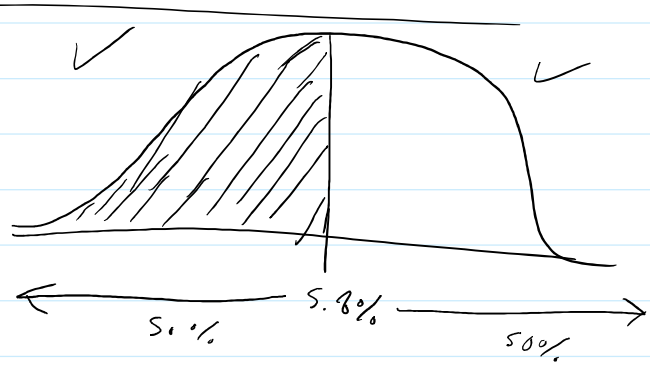
$$R_P = w_A \cdot R_A + w_G \cdot R_G = 20\% \cdot 5\% + 80\% \cdot 6\% = 5.8\%$$

$$\sigma_P^2 = w_A^2 \sigma_A^2 + w_G^2 \sigma_G^2 + 2 \cdot w_A \cdot w_G \cdot \sigma_{AG}$$

$$= 20\%^2 \cdot 0.0144 + 80\%^2 \cdot 0.0100 + 2 \cdot 20\% \cdot 80\% \cdot (-0.0006)$$



Prob 5



$\sigma_p = 7.11\%$

$R_p = 5.8\%$

$R_A = 5\%$

$RP_A = 4\% = R_A - R_f \Rightarrow R_f = 1\%$

$$\text{Sharpe}_p = \frac{R_p - R_f}{\sigma_p} = \frac{5.8\% - 1\%}{7.11\%} = 0.68$$